

AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT				1. CONTRACT ID CODE J W912QR25BA028_Specs-Q002		PAGE OF PAGES 0002 2			
2. AMENDMENT/MODIFICATION NUMBER 0002			3. EFFECTIVE DATE 26 SEP 2025		4. REQUISITION/PURCHASE REQUISITION NUMBER		5. PROJECT NUMBER (If applicable) 965257		
6. ISSUED BY W072 ENDIST LOUISVILLE KO CONTRACTING DIVISION, 600 DR MARTIN LUTHER KING JR PL LOUISVILLE, KY 40202-2230 UNITED STATES MARCIE BILLOTTO, EMAIL: MARCIE.BILLOTTO@USACE.ARMY.MIL TELEPHONE: (502) 315-6213			CODE W912QR		7. ADMINISTERED BY (If other than Item 6) SCD: PAS:			CODE	
8. NAME AND ADDRESS OF CONTRACTOR (Number, street, county, State and ZIP Code)					(X)		9A. AMENDMENT OF SOLICITATION NUMBER W912QR25BA028		
					☒		9B. DATED (SEE ITEM 11) 22 AUG 2025		
					☐		10A. MODIFICATION OF CONTRACT/ORDER NUMBER		
					☐		10B. DATED (SEE ITEM 13)		
CODE			FACILITY CODE						

11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS

☒ The above numbered solicitation is amended as set forth in Item 14. The hour and date specified for receipt of Offers ☐ is extended. ☒ is not extended.

Offers must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended, by one of the following methods:
 (a) By completing items 8 and 15, and returning 1 copies of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted;
 or (c) By separate letter or electronic communication which includes a reference to the solicitation and amendment numbers. FAILURE OF YOUR ACKNOWLEDGMENT TO BE
 RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER.
 If by virtue of this amendment you desire to change an offer already submitted, such change may be made by letter or electronic communication, provided each letter or electronic
 communication makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.

12. ACCOUNTING AND APPROPRIATION DATA (If required)
 SEE SECTION G - CONTRACT ADMINISTRATION DATA

**13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACTS/ORDERS.
IT MODIFIES THE CONTRACT/ORDER NUMBER AS DESCRIBED IN ITEM 14.**

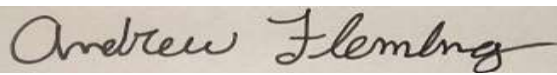
CHECK ONE	A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: (Specify authority) THE CHANGES SET FORTH IN ITEM 14 ARE MADE IN THE CONTRACT ORDER NUMBER IN ITEM 10A.
☐	
☐	B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT THE ADMINISTRATIVE CHANGES (such as changes in paying office, appropriation data, etc.) SET FORTH IN ITEM 14, PURSUANT TO THE AUTHORITY OF FAR 43.103(b).
☐	C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:
☐	D. OTHER (Specify type of modification and authority)

E. IMPORTANT: Contractor ☐ is not ☐ is required to sign this document and return _____ copies to the issuing office.

14. DESCRIPTION OF AMENDMENT/MODIFICATION (Organized by UCF section headings, including solicitation/contract subject matter where feasible.)

THIS AMENDMENT OUTLINES THE FOLLOWING SPECIFICATION SECTIONS HAVE BEEN REVISED AND REPLACED IN THEIR ENTIRETY:
 35 05 40.14 10HYDRAULIC POWER SYSTEMS FOR CIVIL WORKS STRUCTURES
 ALL OTHER TERMS AND CONDITIONS REMAIN UNCHANGED.

Except as provided herein, all terms and conditions of the document referenced in Item 9A or 10A, as heretofore changed, remains unchanged and in full force and effect.

15A. NAME AND TITLE OF SIGNER (Type or print)		16A. NAME AND TITLE OF CONTRACTING OFFICER (Type or print) ANDREW FLEMING	
15B. CONTRACTOR/OFFEROR	15C. DATE SIGNED	16B. UNITED STATES OF AMERICA 	16C. DATE SIGNED 26 SEP 2025
(Signature of person authorized to sign)		(Signature of Contracting Officer)	

SECTION SF 30 BLOCK 14 CONTINUATION PAGE

SUMMARY OF CHANGES

515804 Multilevel Intake and Hydraulic System Repairs
Patoka Lake, IN

SECTION 35 05 40.14 10

HYDRAULIC POWER SYSTEMS FOR CIVIL WORKS STRUCTURES

08/21

Amendment 0002

PART 1 GENERAL

1.1 SUMMARY

The work covered by this section consists of detailed requirements for removal and furnish, testing, delivery, and installation of the replacement part of the hydraulic systems.

1.2 SPECIAL COORDINATION

In addition to the operability and acceptance requirements contained herein, the Contractor must complete the procedures indicated in SECTION 00 80 00.00 06 SPECIAL PROVISIONS PARA in order for work to be fully accepted by the Government.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.11	(2021) Forged Fittings, Socket-Welding and Threaded
ASME B31.1	(2022) Power Piping
ASME B40.100	(2022) Pressure Gauges and Gauge Attachments
ASME BPVC SEC IX	(2023) BPVC Section IX-Welding, Brazing and Fusing Qualifications
ASME BPVC SEC VIII D1	(2023) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 2 2022) Structural Welding Code - Steel
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ASTM INTERNATIONAL (ASTM)

ASTM A182/A182M	(2024) Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service
ASTM A193/A193M	(2024) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and

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Other Special Purpose Applications

ASTM A194/A194M	(2023) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
ASTM A269/A269M	(2024) Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A312/A312M	(2024) Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
ASTM D3951	(2018; R 2023) Standard Practice for Commercial Packaging
ASTM F3125/F3125M	(2023) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 1219-1	(2016) Fluid Power Systems and Components Graphic Symbols and Circuit Diagrams - Part 1: Graphic Symbols for Conventional Use and Data-Processing Applications
ISO 1219-2	(2012) Fluid Power Systems and Components Graphic Symbols and Circuit Diagrams - Part 2: Circuit Diagrams
ISO 4021	(1992) Hydraulic Fluid Power - Particulate Contamination Analysis - Extraction of Fluid Samples from Lines of an Operating System
ISO 4406	(2021) Hydraulic Fluid Power - Fluids - Method for Coding the Level of Contamination by Solid Particles
ISO 4407	(2002) Hydraulic Fluid Power - Fluid Contamination - Determination of Particulate Contamination by the Counting Method Using an Optical Microscope
ISO 4413	(2010) Hydraulic Fluid Power - General Rules and Safety Requirements for Systems and Their Components
ISO 11500	(2022) Hydraulic Fluid Power - Determination of the Particulate Contamination Level of a Liquid Sample by Automatic Particle Counting Using the

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Light-Extinction Principle - Second
Addition

ISO 15614-1 (2017) Specification and Qualification of
Welding Procedures for Metallic Materials
- Welding Procedure test - Part 1: Arc and
Gas Welding of Steels and Arc Welding of
Nickel and Nickel Alloys - Second Edition;
Corrected Version 10/2017

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS
INDUSTRY (MSS)

MSS SP-58 (2018) Pipe Hangers and Supports -
Materials, Design and Manufacture,
Selection, Application, and Installation

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE J514 (2012) Hydraulic Tube Fittings

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 200-1-1 (1994) Environmental Quality -- Validation
of Analytical Chemistry Laboratories

EM 200-1-6 (1997) Environmental Quality -- Chemical
Quality Assurance for Hazardous, Toxic and
Radioactive Waste (HTRW) Projects

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

EM 1110-2-1424 (2016) Engineering and Design --
Lubricants and Hydraulic Fluids

1.4 DESIGN AND PERFORMANCE REQUIREMENTS

The Contract drawings indicate the general arrangement of the hydraulic control station for operation of the control gates, and multi-level gates. Hydraulic pipe/ tube layout. Furnish the detailed design in accordance with the following design criteria. Furnish the detailed design in accordance with ISO 4413, and the following design criteria. Identify design and dimensional changes necessary to satisfy the principal design parameters identified in the paragraph "Design Parameters" as variations in the submittals. Dimensions submitted that differ from those indicated and not otherwise constrained by the Design Parameters and physical limitations of installation are subject to approval. Also submit design computations for all extension of design items. Contractor designed items must be designed to withstand the maximum loading with a minimum safety factor of 5 with regard to the ultimate tensile, compressive, and shear strength and a minimum safety factor of 2 with regard to the tensile, compressive, and yield strength.

The hydraulic equipment supplier is responsible for selecting, furnishing, removing existing valves and tubes, installing and commissioning the up-graded hydraulic system. The hydraulic system designer must be a registered Professional Engineer with at least 10 years of experience designing such equipment. The electrical controls supplier must be a

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subcontractor of the hydraulic system supplier. Mechanical and structural design criteria must follow USACE EM 1110-2-2610 and ETL 1110-2-584.

1.4.1 Design Parameters

The principal design parameters for the hydraulic power system are as follows and were developed at the minimum operating temperature and utilizing the biodegradable hydraulic fluid. as specified. Contractor must determine final values once Valve flow rates will be determined by the Contractor to achieve the gate operation speeds shown below. Changes can be submitted to the Government for approval.

System pressure

- a. Maximum system pressure 3000 psi
- b. Working system pressure 1500 psi
- c. Design system pressure 7500 psi

1.4.2 Connections

1.4.2.1 Shop Connections

Design shop connections for assembly by means of welding or by bolting.

1.4.2.2 Welded Connections

Design welded connections in accordance with AWS D1.1/D1.1M except that provisions for repeated stress is not required. Weld hydraulic cylinders in accordance with ASME BPVC SEC VIII D1, ISO 15614-1. Weld piping in accordance with ASME B31.1. Qualify procedures and welders in accordance with ASME BPVC SEC IX. Welding procedures qualified by others, and welders and welding operators qualified by another employer may be accepted as permitted by ASME B31.1. U-stamp requirements per ASME BPVC are waived. Notify the Contracting Officer within 24 hours in advance of tests and perform the tests at the work site if practicable. Apply the welder's or welding operator's assigned symbol near each weld made as a permanent record. Submit a copy of qualified procedures and a list of names and identification symbols of qualified welders and welding operators.

1.4.2.3 Structural Bolted Connections

Make structural bolted connections carrying primary loads with ASTM F3125/F3125M, Grade A325A490 bolts or ISO equivalent standards as approved.

1.5 SUBMITTALS

Government approval/acceptance by the Resident Office (RO) is required for submittals with a "G" designation; submittals not having a "G" designation are for information only (FI0) or as otherwise designated. When used, a designation following the "G" designation identifies an additional office that will review the submittal for the Government. The following shall be submitted in accordance with LRL Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Installation Procedures; G, RO

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Piping Installation; G, R0

Field Pressure Testing Plan; G, R0

Field Functional Testing Plan; G, R0

Field Acceptance Testing Plan; G, R0

Installation Plan and Procedures; G, R0

SD-02 Shop Drawings

Schematic Drawings and Data; G, D0

SD-03 Product Data

Materials and Mechanical Equipment; G, D0, R0

Standard Products; G, R0

Electrical Equipment; G, R0

Design and Performance Requirements; G, D0

Erecting Engineer Qualifications; G

Cleaning and Flushing; G, R0

SD-06 Test Reports

Shop Testing Report; G, R0

Field Pressure Testing Report; G, R0

Field Functional Testing Report; G, R0

Field Acceptance Testing Report; G, D0

Shop Oil Testing Reports; G,

Field Oil Testing Reports; G,

SD-07 Certificates

Notification Of Shop Testing; G

SD-10 Operation and Maintenance Data

Operation and Maintenance; G, D0

1.6 SCHEMATIC DRAWINGS AND DATA

1.6.1 Shop Drawings

Include fabrication, shop assembly, delivery, and field installation drawings in the detailed shop drawings. Detail any component part of fabricated items omitted on the shop drawings. If departures from the Contract drawings are deemed necessary by the Contractor, submit details of such departures, including changes in related portions of the project and reasons, with the shop drawings. Provide fabrication and assembly

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drawings to or 1-1/2 inch = 1 foot-0 inch or larger scale.

1.6.2 Fabrication Drawings and Assembly Details

Provide fabrication drawings for all mechanical and structural parts and components, except those of standard design and manufacture. Include in fabrication drawings complete detailing, materials of construction, tolerances, machined surface finishes, connections, and weld details with annotation to differentiate shop welds from field welds.

1.6.3 Piping Drawings

Provide piping drawings showing the hydraulic control station and multi level gate area to include all new work in schematic format identifying all items of equipment incorporated in the system. Include details of all pipe and valve supports.

1.6.4 Shop Assembly Drawings

Provide shop assembly drawings with details for connecting the adjoining fabricated components in the shop to ensure satisfactory field installation.

1.6.5 Hydraulic Schematic

Provide complete hydraulic schematics hydraulic control station and multi level gate area to include all new work in accordance with ISO 1219-1 and ISO 1219-2. Schematics shall include schematic piping, hoses and other component layouts symbolically indicating, using standard system symbols, all system piping, and other components, including their sizes, materials, heights, spacing, and locations. Both new and refurbished components shall be shown on the schematic with any interfaces to existing equipment noted. All set point and size parameters shall be indicated for each component. Component support types and locations, anchor points versus sliding supports, flex joints, and seismic supports shall also be shown. Include proposed pertinent installation details. Required capacities and system pressures as well as direction of system flow or motion shall be indicated. Each drawing shall include a legend of symbols and bill of materials (BOM). BOM to include name of component, component drawing number or reference drawing, and part number. If hydraulic schematic is spilt between drawings or connect to a sub-contractor supplier drawing the connection points must be clearly outlined on both the schematic and other drawing, utilizing identical nomenclature.

1.6.6 Product Data

Include performance data and curves for all types of valves. Provide catalog cuts and outline dimensions for all valves, valve controls and other accessories.

1.6.7 Field Installation Procedures

Provide field installation drawings with a detailed description of the field installation procedures. Include the location and method of support of installation and handling equipment, the provisions to be taken to protect concrete and other work during installation, the method of maintaining components in correct alignment, and the methods for installing other appurtenant items. Include dates and schedule of work. Submit methods and procedures to accelerate the sequence of work. All new

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work in accordance with EM 385-1-1

1.6.8 Design and Performance Requirements

Provide design computations for all items which are designed by the Contractor. Include computerized simulations of the complete and proposed hydraulic system. All new work in accordance with EM 1110-2-1424, EM 1110-2-2610

1.6.9 Cleaning and Flushing Procedures and Results

Submit detailed procedure for the cleaning and flushing of hydraulic cylinders, piping, and power unit as outlined in paragraph "CLEANING AND FLUSHING," not less than 30 calendar days before start of operations. Provide test reports of the results of the flushing and of the oil used to fill the system. All new work in accordance with EM 385-1-1, EM 200-1-6, EM 1110-2-1424, and EM 200-1-1

1.6.10 Erecting Engineer Qualifications

Provide a resume for the on-site erecting engineer with details on experience and background in similar installations.

1.7 WARRANTY

Guarantee all equipment for a period of 2 years from the date of acceptance. Guarantee replacement parts for 2 years from date of replacement. Provide warranty against defective materials, design, and workmanship. In cases where the equipment manufacturer's advertised minimum guarantee is in excess of 2 years, it remains in force for its full period. Upon receipt of notice from the Government of failure of any of the parts during the warranty period, provide new replacement parts promptly at no additional cost to the Government.

1.8 QUALITY CONTROL

Establish and maintain quality control for operations to assure compliance with Contract requirements and maintain records of quality control for all materials, fabricated parts, equipment, and construction operations. In addition, establish and maintain surveillance for quality control over sub-contractors, suppliers, or manufacturers. The quality control includes, but is not limited to, the following:

- a. Materials and workmanship.
- b. Installation of the piping complete.
- c. Cleaning and flushing.
- d. Field erection and tests.
- e. Damage and defects.

Submit a copy of these records and tests, as well as the records of corrective action taken, to the Government.

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PART 2 PRODUCTS

2.1 MATERIALS AND MECHANICAL EQUIPMENT

2.1.1 General

Provide materials and mechanical equipment in accordance with the requirements as indicated or specified, and if not specified, furnish materials and mechanical equipment of the best commercial grade quality suited to the intended use and as approved. All valves and similar items and/or accessories, of the same type and size, to be the products of the same manufacturer, unless otherwise approved. Permanently display the manufacturer's name, address, and catalog number on a nameplate securely attached to each major item of equipment.

Submit data specifications product manuals and assembly drawings showing sizes, ratings, parts and material lists, overall dimensions, and mounting dimensions with the product data.

2.1.2 Standard Products

Where items are referred to hereinafter as "similar and equal to" a particular manufacturer's product, such references have been made merely as a convenient method of indicating the type of material or equipment required, with no intention of asserting superiority thereof. The standard product of any reputable manufacturer regularly engaged in the commercial production for at least 2 years prior to this solicitation of the type and quality of material or equipment referred to will not be excluded on the basis of minor differences, provided essential requirements of the specifications relative to materials, capacity, and performance are met. Furnish performance capacities and other pertinent information concerning the manufacturer's "equal to" standard products intended for incorporation in the work. "Equal to" standard products installed or used without such approval are at the risk of subsequent rejection.

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2.2 PIPE, TUBE, HOSE, AND APPURTENANCES

2.2.1 Piping

2.2.1.1 Pipe - General

Design piping for a working pressure of 3000 psi. Weld piping in accordance with ASME B31.1. Qualify procedures and welders in accordance with ASME BPVC SEC IX. Welding procedures qualified by others, and welders and welding operators qualified by another employer may be accepted as permitted by ASME B31.1. Provide pipe and fittings in accordance with ASME B31.1 and subsequent addenda unless otherwise specified or indicated.

2.2.1.2 Pipe/ Tube

Use 316 seamless stainless steel tube linesnd ASTM A269 standards stainless stee in accordance with ASTM A312/A312M, and ASTM A269/A269M.

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2.2.1.3 Tube Fittings

All tube fittings shall be swaged or flare type. Bite-type shall not be used. Fitting / hose assemblies shall be compliant with SAE J514.

2.2.1.4 Pipe Fittings

Use socket welding type pipe fittings in accordance with ASME B16.11 and made of stainless steel in accordance with ASTM A182/A182M, Grade F304. Provide pressure rated to 3000 psi. Flanges in accordance with ASTM A182/A182M with the grade suitable for the pipe to which attached. Also conform threaded fittings to the above, but use only where absolutely necessary for the application.

2.2.1.5 Unions

Provide O-ring type unions, made of stainless steel rated to 3000 psi. The Contractor may, as an option, substitute four bolt split flanges with Buna N O-rings for the unions.

2.2.2 [Enter Appropriate Subpart Title Here] 2.2.2.1 [Enter Appropriate Subpart Title Here]

2.2.3 Valves

Provide a minimum pressure rating of 3000 psi unless stated otherwise. Valves to be specifically designed and rated for hydraulic system applications. Provide nameplates for all valves and submit for approval. Nameplates to clearly indicate the function of each device and indicate the condition established for each. Machine engrave lettering on nameplates on steel plate.

2.2.3.1 Shut-Off Valves

Provide SAE Code 61 flanged Shut-Off Valves. stainless steel, lever operated, ball type. Shut-off valves at each cylinder for the main supply and return lines to be match the existing line size, rated for a working pressure of 3000 psi, double acting ball type. Shut-off valves for all other lines to be ball type, match the line size, and have a maximum allowable working pressure of 2000 psi. The valve ends to have connections and drilled to receive SAE Code 61 flanges. Provide a removable operating lever for each valve. Valves to be specifically designed and rated for hydraulic system applications. Provide with stainless steel valve trim including handles. The valves to have replaceable seats and be repairable without disturbing the welded connections.

2.2.3.2 Manual Control Valves

2.2.3.2.1 Flow

Flow control valves to be line mounted. The valves to be, free flowing in one direction, and adjustable. The valves to be capable of being locked in position to prevent an unintentional adjustment. The flow rating to be determined by the Contractor in accordance with the gate speed design criteria stated in paragraph "Design Parameters".

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2.2.3.2.2 Manual Four-Way Directional Control Valves

The rotary shear seal type, open or closed center, and spring centered. Valves to be lever operated three position, subplate mounted at the hydraulic control stations. The flow rating to be determined by the Contractor in accordance with the design criteria stated in paragraph "Design Parameters."

2.2.3.3 Supply Spring Loaded Check Valves

Provide supply spring loaded check valves of stainless steel construction, the ball or poppet type with a body designed for high shock and 3000 psi service.

2.2.3.4 Return Spring Loaded Check Valves

Construct the return spring loaded check valves of stainless steel, the ball or poppet type with a body designed for 3000 psi service. Cracking pressure to be 500 psi.

2.2.3.5 Bleeder Valves

Provide or 1/4 inch, stainless steel construction, wrench operated bleeder valves.

2.2.3.6 Pressure Snubbers

Provide stainless steel pressure snubbers for all pressure gauges and pressure switches to protect against shock and provide more stable instrument operation.

2.2.3.7 Counterbalance Valve

Install in the oil line to the bottom side of the hoist piston as indicated to balance the load being held by the cylinder. The valve to be directly operated, internally drained, and adjustable for operating over a pressure range of 500 to 2500 psi. Design the valve for a system operating pressure of 1500 psi, but pressure rated to 3000 psi. Provide the valve capacity rating to match the rating of the existings valves. Permit unrestrained flow to the underside of the hoist piston and retain pressure in the hoist cylinder in the amount of the valve's pressure adjustment. Each valve to have an adjustable flow control valve in the pilot pressure line. The counterbalance valve to be factory set in accordance with the settings as indicated.

2.2.4 Pipe Hangers and Supports

Locate all pipe support devices at intervals no greater than or 6 feet between centerlines of adjacent supports, except as modified as specified. Install support devices on both sides of a bend within four nominal pipe or tube diameters of the bend location. Furnish all supports, hangers, sleeves and brackets complete with compatible mounting hardware and appurtenances. Conventional pipe hangers and support in accordance with MSS SP-58 type as required. Provide stainless steel pipe supports and hangers. Provide special hangers and anchors as indicated. Construct tube supports of stainless steel hardware with polypropylene support halves. Provide each tube support with all mounting hardware required to connect with the appropriate anchorage system.

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2.2.5 Sleeves and Wall Brackets

Fabricate sleeves and wall brackets of stainless steel as indicated.

2.2.6 Pipe Penetration Seals

Provide modular mechanical type pipe penetration seals, consisting of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe and opening. Assemble links to form a continuous rubber belt around the pipe with a pressure plate under each bolthead and nut. Tightening of the bolts to cause the rubber sealing elements to expand and provide a watertight seal.

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2.3 INSTRUMENTS AND APPURTENANCES

2.3.1 Pressure Gauges

Provide in accordance with ASME B40.100, with a black enameled metal case, a or 4-1/2 inch dial, and a stainless steel Bourdon tube. The scale range of the gauge to be approximately 150 percent of the maximum pressure of the line in which installed. Provide safety type gauges with solid fronts and blowout backs. Provide each gauge with a pressure snubber. Mount gauge as indicated. Provide bottom tap gauges and gauge lines in horizontal pressure lines.

2.4 HYDRAULIC FLUID

2.4.1 Petroleum Based

Use the hydraulic fluid during shop testing, flush the system after installation, and to fill the complete hydraulic system to be AW32 Petroleum Corporation's ISO 32 Hydraulic Fluid hydraulic oil which has a high viscosity index, low pour point, and antifoam properties, See section 01 02 10.00 06 for AW32 hydraulic fluid Specifications. Filter fresh hydraulic fluid through a 10 µm filter before it is added to the system. Introduce clean and fresh hydraulic fluid 10 µm filter before it is added to the system. Provide letters of assurance from the valve manufacturers that the oil provided is satisfactory for use in their equipment in this application. Supply all oil and furnish additional one 55 gallon containers to the Government for a reserve supply.

2.5 FASTENERS

2.5.1 Stainless Steel Bolts and Nuts

Provide in accordance with ASTM A193/A193M, Grade B8, with ASTM A194/A194M, Grade 8 nuts.

PART 3 EXECUTION

3.1 EXAMINATION

After visiting the site and becoming thoroughly familiar with all details of the work and working conditions, verify dimensions in the field, and then advise the Contracting Officer of any discrepancies prior to performing any work. The Contractor is specifically responsible for the

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coordination and proper relation of the contracted work to the structure and work of all trades.

3.2 SHOP FABRICATION

3.2.1 Painting

Shop prime and coat all exposed exterior surfaces of assemblies and equipment except stainless steel, synthetic rubber, and plastic. HYDRAULIC STRUCTURES unless the equipment is given a standard factory finish as specified. Insofar as is practicable, apply the complete coating system to individual components and items before assembly to ensure complete coverage and maximum protection against corrosion. Paint shop assembled equipment prior to any storage or shipment. Do not paint aluminum, stainless steel, non-ferrous surfaces, or machined surfaces requiring field assembly, except as approved. Paint standard manufactured equipment, valves, in accordance with the manufacturer's standard practice for high humidity service, subject to approval. Repair chips, scratches, and other damage to shop-applied painted surfaces in the field.

3.3 SHOP ASSEMBLY

3.3.1 Protection During Assembly

During assembly securely cover all openings to avoid the entrance of abrasives, dirt, metal chips, and other foreign materials into the hydraulic system through open ends of piping, tubing, and ports of the components.

3.3.2 Cleaning and Protection During Assembly

During assembly, ensure all components are free of abrasives, dirt, metal chips and other foreign materials. Clean components as needed to remove foreign materials. Securely cover openings to prevent recontamination.

3.3.3 Flushing

After completing assembly, flush each cylinder and HPU. Use the same hydraulic fluid for flushing as approved for the final filling. Filter oil using a 10 µm filter. Perform flushing until the contamination level is 18/16/13 in accordance with ISO 4406.

Take three 500 milliliter samples at approved locations in accordance with ISO 4021. Perform particle counting on each sample in accordance with ISO 11500 or ISO 4407 by an approved independent test laboratory. Water content of each sample to be below 200 ppm. Reclean if any sample does not comply with the permissible contamination limits, and reinspect. Submit shop oil testing reports.

3.4 SHOP TESTING

3.4.1 Draining of Fluid

Upon successful completion of flushing and testing operation, drain the flushing fluid from the equipment.

3.4.2 Testing Plan and Procedures

Prepare and submit a testing plan and procedure prior to the start of

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testing. Testing activities are not allowed until the testing plan is approved. Submit the testing plan a minimum of 30 days prior to the start of testing. Include the following elements, as a minimum, in the testing plan:

- a. A step-by-step, repeatable procedure for each test to be performed, including but not limited the tests detailed above.
- b. Diagrams or drawings of test setups showing orientation of equipment during testing and installation of instrumentation.
- c. List of instrumentation that will be used to take measurements.
- d. Sample data collection sheets, clearly identifying what is to be recorded, to what accuracy, and the planned outcome (such as value range, threshold, or condition).

3.4.3 Testing Report

After completion of each test or group of tests, submit a test report. Include the following elements, as a minimum, in the testing plan:

- a. Date(s) testing occurred
- b. Identification of which equipment was tested
- c. Completed data collection sheets.
- d. Provide records of all adjustments and final settings for all hydraulic components in the system. This includes pump flow rates, operating times, relief settings, counterbalance valve adjustments, operating pressures at both the hydraulic power units and cylinder, flow level adjustments, filter differential switch settings, and other parameters.
- d. Post-processed output from any data collection equipment
- e. Identification of a successful test. If any test stoppages or failures occurred, identify them and the measures taken to correct the deficiencies.
- f. Current, non-expired calibration certificates for all instrumentation and measurement devices used during testing.

3.5 SHIPPING, HANDLING, DELIVERY, AND STORAGE

3.5.1 Packaging

Any delivery of components must be coordinated by the Contractor and noted in a submittal. The subassemblies are defined as the following:

- a. Piping assemblies.
- b. Control consoles.
- c. Valves.

Provide all subassemblies and spare equipment and parts with adequate protective pads, supports, and blocking, and securely restrained to

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prevent distortion or damage to the painted surfaces in transit. Any loss or damage during shipment, including damage to the painted surfaces, will be the Contractor's responsibility; replace or repair without any cost to the Government. Pack all accessories and spare parts separately in containers plainly marked "ACCESSORIES ONLY" or "SPARE PARTS ONLY." Place a packing list, listing the contents of each container, in a moisture-proof envelope and securely fasten to the outside of the container. Standard commercial packaging in accordance with ASTM D3951 will be acceptable except where a different method or standard of packaging is specified.

3.5.2 Shipping, Preservation, and Storage

Packing, crating, cradles, and other packing materials necessary to ensure safe shipment are the responsibility of the Contractor and become the property of the Government upon delivery of the equipment. Then securely cap remaining ports with blank flanges to prevent the entrance of foreign matter. Adequately protect machined surfaces from corrosion and physical damage. Protect equipment delivered and placed in storage from the weather, humidity, temperature variation, dirt and dust, or other contaminants.

3.6 ON-SITE INSTALLATION

Install the equipment specified and as shown to complete the hydraulic control stations for operation of the control gates. Install hydraulic components in accordance with the manufacturer's written instructions and under the direction of the erection engineer or manufacturer's representative. Install complete units or assemblies without disassembly. Provide necessary supports for all appurtenances, pumps, motors, and other equipment or components as indicated. Submit an installation plan and procedures.

3.6.1 Erection Engineer

Obtain the services of an experienced erection engineer who is regularly employed by the hydraulic cylinder/power unit manufacturer to supervise the installation, start-up, adjustment and operation, and testing of the equipment provided. The erection engineer to furnish a signed statement stating that the final installation and start-up of the hydraulic power system has been inspected, witnessed, and complies fully with the manufacturer's warranty requirements. Following completion of the work the erecting engineer to instruct the Contracting Officer in the operation and maintenance of the system. These field instructions to cover all items contained in the bound instructions. Do not conduct instruction until Operation and Maintenance Manuals are approved.

3.6.2 Filling And Bleeding The System

With all hydraulic equipment installed and all cleaning and flushing complete filling of the system may begin. The cleanliness of the oil used to fill the system in accordance with the requirements for system cleanliness. Open the by-pass ball valves at the cylinders. Fill each hydraulic power unit and pump oil into the system through a 10 µm filter and keep adding oil as long as the level continues to drop. Cease pumping when the oil level no longer drops. Install by-pass piping and repeat the procedure to fill the drain/siphon line. Take care to expel as much air as possible from the piping and cylinders during the initial filling. Utilize piping vents and drains as much as possible to expel air from the

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system. Each power circuit pipe for the cylinders to have this procedure performed to fill the respective piping. Actuate cylinders by shifting spool on the four-way valve back and forth and bleeding air from ports provided on cylinders. Continue procedure for all hydraulic power units until all cylinders have been bled. Ball valves in tank lines may be closed during bleeding to prevent introduction of air into tank lines. After all the cylinders for the machinery have been filled and bled of air, fill the hydraulic power unit tanks with oil. The oil level with the control gates open to be above the "low level" shut-off. The system, once filled, to be bled of air, operated, and periodically bled during the first week of operation to remove any air entrained in the system.

3.6.3 Used Hydraulic Oil

Contractor is responsible for the containment, transfer, removal, and disposal of all used hydraulic oil. Used hydraulic oil includes any oil in existing components being removed or refurbished, and any generated used oil from flushing, draining, or bleeding activities as identified in this section. Containment, transfer, removal, and disposal procedures for the used oil must be detailed in the applicable removal and installation procedures.

3.6.4 Power Piping and Hoses

The general arrangement of the hydraulic piping and hoses is as indicated. Any changes to the arrangement necessary to facilitate the installation and proper functioning of the system may be made subject to the approval of the Contracting Officer.

3.6.4.1 Piping Installation

Install the system complete including all necessary valves, fittings, and pipe accessories. All joints to be tight and successfully pass the test as specified. Submit details of pipe supports and anchors not indicated. Adequately support all lines at intervals not greater than or 10 feet or as otherwise indicated. Install hangers and supports using machine bolts and masonry anchors caulked in drilled holes in the masonry or by using machine bolts and expansion shields. Power piping must not be used to support other equipment. Other equipment can use the same standoff or hanger mounting point as one to support the piping, but the pipe itself must not provide the support of the equipment's load. Spacers and guides can be used if no load is transferred. Ream and remove burrs from all cut ends of pipe. Remove metal particles from the reaming operations and thoroughly clean ends of pipe before proceeding with the work. Tightly plug all piping at all times except when work is being performed on a pipe. Clean the pipe sleeves, recesses and trenches of all debris and thoroughly wash with a high pressure stream of water before any piping is installed. Drilling, chipping, or grinding of concrete in close proximity of any piping being installed is not permitted. Repeat washing out of the sleeves, recesses and trenches from time to time as necessary. Store valves in a clean, dry place and protect against moisture. Do not install valves in the system prior to 4 months before the system is to be filled with hydraulic oil. Install pipes passing through masonry in pipe sleeves as indicated. Mitering of joints for elbows and notching of straight runs of pipe for tees is not permitted.

3.6.4.2 Support of Valves with Manual Operators

Valves that operate through a manual operator (such as a handwheel or

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handlever)connecting to hydraulic tubing must have a hanger or support on each side of the valve within 1 inch of the connection. Such valves include block, ball and needle valves. The hangers or supports must translate the load from the use of the manual operator away from the connection of the valve and tubing. Valves used with hydraulic piping larger than 2 inches in diameter may not require this additional support if the manual operator can be used without induction detrimental induction of forces into the assembly.

3.6.4.3 Identification of Piping and Valves

Identify all pipe, tubing, valves, fittings, as required, and hydraulic power equipment, located within the trenches, machine rooms, and machinery recesses. Use No. 20 gauge brass tags for tagging, with the proper identification symbol stamped into the metal. Use piping symbols as indicated and stamp with or 3/4-inch high lettering. Attach tags to piping and valving by means of No. 12 gauge copper wire. Install pipe markers at intervals of no greater than or 15 feet, except as approved.

- a. Provide a numerical identification tag on each valve, coded such that no other valve in a connected hydraulic power system has the same number. Provide each power circuit valve with an additional tag, which indicates the valve's function.
- b. Identify all other valves at the port location, on the connecting piping.
- c. No tags are needed on tubing below EL 549.0.
- d. Identify all ball, bleed, and globe valves with a "normally closed" or "normally open" legend.
- e. Identify all gauge mounts and pressure transducer mounts with numerical tags, as well as any working or instruction tags required for safe operation.
- f. Provide a warning tag and an instruction tag, or tags as required, at each return filter assembly to indicate the safe, approved procedures for cartridge replacement and by-pass operation.

3.7 CLEANING AND FLUSHING

During assembly, securely cover all openings to avoid the entrance of abrasives, dirt, metal chips, and other foreign materials into the hydraulic system through open ends of piping, tubing, and ports of the components. Use the same hydraulic fluid for flushing as approved for final filling. Include a detailed description of the equipment, materials, hydraulic fluid, temperatures, and duration of each phase of the flushing in the procedures. Clean the system of particles so that the contamination level is below 18/16/13 in accordance with ISO 4406. Take three 500 milliliter samples at approved locations in accordance with ISO 4021. Perform particle counting on each sample in accordance with ISO 11500 or ISO 4407 by an approved independent test laboratory. Water content of each sample to be below 200 ppm. Reclean if any sample does not comply with the permissible contamination limits, and reinspect. When flushing is completed, drain the system and then fill with the specified hydraulic fluid introduced through a 10 micron filter.

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3.7.1 Flushing Piping

Flush all hydraulic piping before installation. Install by-pass loops of piping in place of cylinders and the power units. Circulate hydraulic fluid through each and every pipe unit until returning oil meets the requirement for system cleanliness. Sequence flushing so that all piping is flushed in both directions. The flow capacity of the flushing system to produce a minimum velocity of or 15 feet per second in all piping. Provide means to verify the flow during the flushing operation.

3.8 FIELD TESTS AND INSPECTIONS

3.8.1 General Requirements - Field Testing

The following requirements apply to all testing activities:

- a. Notify the Contracting Officer at least 14 calendar days before any field testing is to be conducted.
- b. Conduct testing in the presence of the Contracting Officer unless waived in writing.
- c. Conduct testing under the direction of the erection engineer or manufacturer's representative.
- d. Constantly monitor details of all operations for signs of impending trouble or leakage and make corrections as necessary to prevent damage to the equipment.
- e. Immediately correct any deficiency or maladjustment disclosed by the tests and repeat the test until satisfactory results are obtained. No subsequent tests will be permitted until all preceding tests have been completed satisfactorily.
- f. No subsequent tests will be permitted until all preceding tests have been completed satisfactorily.

3.8.2 Field Pressure Testing

3.8.2.1 Pressure Testing Requirements

Perform a hydrostatic pressure test for all new installed piping and equipment. Pressure testing may be split up into distinct and separate portions as determined by the Contractor. Perform the pressure testing in accordance with the following requirements:

- a. Use the hydraulic fluid specified in this SECTION as the test fluid.
- b. Use a test pressure of 3000 psi.
- c. Perform each test for a minimum of 1 hour and additional time as needed to verify no leakage.
- d. Inspect all joints, seals, packing, valves, and other areas that are possible leak points. Carefully examine welded, flanged, flared, and threaded connections and wipe for leakage, also inspect lines for evidence of deflection caused by inadequate anchorage.
- e. The criteria for a successful test is zero leaks, zero drop in pressure,

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and no deflection of piping or equipment.

- f. Take means to protect other portions of the system not being tested from becoming inadvertently pressurized during testing.

3.8.2.2 Field Pressure Testing Plan

Prepare and submit a pressure testing plan a minimum of 45 days prior to the start of testing. The plan must be approved prior to the start of testing. Include the following elements, at a minimum, in the plan:

- a. A step-by-step, repeatable procedure for performing the pressure testing. Address how challenging to reach portions of the equipment will be accessed for inspection during the test. Address installation details of temporary instruments and measuring devices.
- b. List of equipment and instrumentation to be used during the test.
- c. Spill protection, controls and counter measures plan to address plans and response to any leaking oil.
- d. Sample data recording sheets.

3.8.2.3 Field Pressure Testing Report

For each pressure test performed, submit a test report. Include the following elements, at a minimum, in the report:

- a. Date(s) testing took place
- b. Identification of the equipment or systems of equipment tested and covered by the test report.
- c. Completed data recording sheets with the results from testing.
- d. Short narrative or statement confirming successful completion of the test.
- e. If issues were discovered during testing and any testing had to be repeated, provide a narrative describing the issues and the measures taken to correct the issues.
- f. Current, non-expired calibration certificates for all instrumentation and measurement devices used during testing.

3.8.3 Field Functional Testing

3.8.3.1 Control Console Functional Testing Requirements

For the control console:

- a. verify the correct operation of each of the devices on the control console, including but not limited to: manual shut-off valves, manual control valves, manual flow control valves, solenoid operated valves, check valves, pressure gauges.
- b. Test all automated control systems and devices.
- c. Functional testing sequence specific to the system.

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3.8.3.2 Field Functional Testing Plan

Prepare and submit a functional testing plan. Submit plan a minimum of 45 days prior to the start of testing. Plan must be approved by the Government prior to the start of testing. Include the following elements in the plan at a minimum:

- a. A step-by-step, repeatable procedure for performing the functional testing. The procedure must individually address testing of the controls and control consoles and the HPU.
- b. List of equipment and instrumentation to be used during the test.
- c. Spill protection, controls and counter measures plan to address plans and response to any leaking oil.
- d. Sample data recording sheets.

3.8.3.3 Field Functional Testing Report

For each functional test performed, submit a test report. Include the following elements, at a minimum, in the report:

- a. Date(s) testing took place
- b. Identification of the equipment or systems of equipment tested and covered by the test report.
- c. Completed data recording sheets with the results from testing.
- d. Short narrative or statement confirming successful completion of the test.
- e. If issues were discovered during testing and any testing had to be repeated, provide a narrative describing the issues and the measures taken to correct the issues.
- f. Current, non-expired calibration certificates for all instrumentation and measurement devices used during testing.

3.8.4 Acceptance Testing

3.8.4.1 General Requirements - Acceptance Testing

Upon completion of all pressure testing and functional testing, perform acceptance testing. Acceptance testing must prove that the complete system, as installed, meets all requirements of the contract. During acceptance testing perform the following:

- a. Monitor and record the following during all acceptance testing. Measurements must be recorded using continuous data logging equipment sampling at a minimum of 10 samples per second:
 - 1) Hydraulic fluid temperature at HPU discharge
 - 2) Flow rate at each cylinder, and at HPU discharge
 - 3) Pressure in supply and return lines at HPU discharge reservoir

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- 4) Speed, temperature, voltage, and amperage of each motor
- 5) Flow Control Valve Settings
- b. During each test operation, inspect the hydraulic lines and components for evidence of leakage.
- c. Check flow control valves and adjust as required to conform to indicated operating time requirements.
- d. Inspect and adjust sequence valves as required to obtain the indicated sequence of operation.
- e. Adjust chokes in pilot circuits of pilot-operated valves to obtain smooth, shock-free operation.
- f. Adjust relief valves and counterbalance valves to the proper pressures as indicated, unless otherwise directed by the Contracting Officer.
- h. Inspect response of components to operation of applicable controls to confirm that all connections have been made properly.
- g. Observe components and for excessive vibration, alignment and operating clearances.
- h. Record observations regarding such events as unusual sounds, malfunctions or difficulties encountered, and adjustments required.

3.8.4.2 Field Acceptance Testing Requirements

Perform the following acceptance tests. Operating tests to cover a period of not less than 4 hours, and conduct all tests at such times as the Contracting Officer may direct. After installation and testing of the hydraulic system has been completed, install a complete set of new and unused filter cartridges.:

- a. After final assembly and installation of the piping/ tubing, and control station, operate each assembly of operating machinery individually as nearly as practicable under its normal operating conditions for a minimum of 5 open/close cycles to demonstrate that each assembly is in proper working order and free from defects of materials, workmanship or alignment.
- b. Upon satisfactory completion of the individual tests, and before final acceptance of the work, conduct, in the presence of the Contracting Officer, over-all testing through a minimum of 10 additional operating cycles, to demonstrate that all machinery has been properly installed and that operation can be effected without interruption.

3.8.4.3 Field Acceptance Testing Plan

Prepare and submit an acceptance testing plan. Submit plan a minimum of 45 days prior to the start of testing. Plan must be approved by the Government prior to the start of testing. Include the following elements in the plan at a minimum:

- a. A step-by-step, repeatable procedure for performing the acceptance testing.

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- b. List of equipment and instrumentation to be used during the test.
- c. Spill protection, controls and counter measures plan to address plans and response to any leaking oil.
- d. Sample data recording sheets.

3.8.4.4 Field Acceptance Testing Report

For each functional test performed, submit a test report. Include the following elements, at a minimum, in the report:

- a. Date(s) testing took place
- b. Identification of the equipment or systems of equipment tested and covered by the test report.
- c. Completed data recording sheets with the results from testing.
- d. Short narrative or statement confirming successful completion of the test.
- e. If issues were discovered during testing and any testing had to be repeated, provide a narrative describing the issues and the measures taken to correct the issues.
- f. Current, non-expired calibration certificates for all instrumentation and measurement devices used during testing.
- g. A written statement that the hydraulic power system has been field tested and meets all operational requirements.

3.8.5 Final Oil Testing

After all field testing is complete, take three 500 milliliter samples at approved locations in accordance with ISO 4021. Perform particle counting on each sample in accordance with ISO 11500 or ISO 4407 by an approved independent test laboratory. Water content of each sample to be below 200 ppm. Reclean if any sample does not comply with the permissible contamination limits, and reinspect. Submit field oil testing reports.

3.9 CLEAN-UP

Keep the work areas clean during installation of the hydraulic system and appurtenances. Upon completion of the installation of the hydraulic system and appurtenances, remove debris and surplus materials resulting from the work.

3.10 OPERATION AND MAINTENANCE

Submit Operation and Maintenance manuals as data packages. Supplemental requirements are as described in this paragraph. Furnish electronic sets of instructions containing the manufacturer's operation and maintenance instructions for each piece of equipment to the Contracting Officer. Permanently bind each set with a hard cover. Furnish one complete set prior to field testing and furnish the remaining sets before the contract is completed. Inscribe the following identification on the covers: "OPERATING AND MAINTENANCE INSTRUCTIONS," title of the project, location

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of the project, the name of the Contractor, and the contract number. Place a flysheet before instructions covering each subject. The instructions include, but are not be limited to, the following:

- a. A system layout drawing showing the piping, valves, and controls.
- b. A system hydraulic schematic indicating final valve pressure settings utilized for fully operational system.
- c. Final as-built drawings of the hydraulic system.
- d. Operating and maintenance instructions.
- e. Manufacturer's bulletins, catalog cuts, and descriptive data.
- f. A written control sequence describing startup, operation, and shutdown. Uniquely identify the control sequence and list the individual components of the hydraulic system. Each component to have a narrative description as to the function, purpose, and limits of adjustment (if any) and method of adjustment for that component.

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-- End of Section --